

Multi-Fidelity Propeller Noise Prediction using a Data-Driven Approach

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Accurate and rapid prediction of propeller aeroacoustic performance is of paramount importance to the design of quiet urban air mobility (UAM) concepts emerging over the last decade. For the broadband noise component, Amiet-type low-fidelity methods are typically used in order to avoid resorting to computationally prohibitive scale-resolving simulations. These methods however, are highly limited in their accuracy and robustness over the range of flow conditions UAM propellers are designed to operate. This paper presents an exploratory effort in developing a multi-fidelity framework to enhance the propeller broadband noise prediction using data-driven approaches. In this framework, a deep neural network machine learning (ML) model is trained in a multi-fidelity manner using transfer learning (TL). The model is first trained using a large number of computationally inexpensive low-fidelity simulations and then enhanced by a small number of high-fidelity aeroacoustic wind tunnel measurements using TL. In the deployment stage, this ML model enables rapid prediction of broadband sound pressure level of isolated 2-bladed propellers between 1kHz and 7kHz at three farfield observer locations given the cross-sectional profile, pitch-to-diameter ratio as well as forward speed and rotational speed of the propeller. Results evaluated based on experimental data from two extra sets of propeller blades, which were withheld from the ML training process, indicated that the multi-fidelity ML model trained with both low- and high-fidelity data based on the TL approach is capable of significantly improving the predictive accuracy of the low-fidelity model. Furthermore, the TL-based multi-fidelity ML model delivered more accurate predictions than the ML models trained solely with limited high-fidelity data.

I. Introduction

Propeller noise is the result of relative motion between the propeller blades and the surrounding air. In particular, it consists of both a tonal and a broadband component. The tonal component is produced by the displacement of air due to the blade motion (thickness noise) and the mean pressure difference between the top and bottom surfaces of the blades as they rotate through the air (loading noise). The broadband component, on the other hand, occurs as a result of turbulence being convected past the trailing-edge of the blade (*i.e.*, self-noise) and the unsteady turbulence interaction of the blades.¹ For an isolated propeller encountering a laminar inflow, the latter can mainly be attributed to blade-wake interaction noise, which is higher than the discrete frequencies of tones and yet lower than the self-noise frequencies.² In conventional scenarios of helicopters and propeller-driven passenger aircraft, tones are considered as the dominant source of noise due to large size of the blades and the high tip Mach number. Since the emergence of next generation Unmanned Aerial Vehicles and Urban Air Mobility (UAM) concepts, propeller-powered systems have garnered renewed interest. As these vehicles operate at a considerably different conditions - lower airspeeds

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and smaller propellers - their noise characteristics are inherently different from those conventional ones. In a recent review by Greenwood *et al.*² on achieving low noise for these vehicles, they highlighted the importance of better understanding and prediction of broadband noise. Indeed, given the non-deterministic nature of the broadband noise and the multiple noise generation mechanisms at play, accurate prediction of the broadband component of the propeller noise has long presented itself as a research challenge, and yet has become increasingly important for the design of future aerial vehicles.

High-fidelity aeroacoustic prediction of propeller broadband noise requires accurate resolution of the pressure fluctuations in the turbulent boundary layer at the blade trailing edge and the turbulent wake. For smaller scale propellers typically used in UAM concepts, key flow features such as laminar-turbulent transition, laminar separation bubble and flow re-attachment have also been shown to act as broadband noise sources. To accurately capture these broadband noise generating mechanisms, scale-resolving simulations such as Lattice-Boltzmann Method (LBM)³ or large-eddy simulation (LES)⁴ are required. However, these methods are computationally intensive and therefore not suitable for many-query tasks such as design optimization. On the other hand, low- and mid-fidelity methods such as those based on Blade Element Momentum Theory (BEMT), vortex-lattice method (VLM) or Reynolds Averaged Navier-Stokes (RANS) simulations tend to perform poorly under off-design conditions and highly twisted blades, and are only capable of capturing the tonal component of the propeller noise, due to their inability to resolve trailing edge turbulent fluctuations responsible for broadband noise generation. Trailing-edge noise models based on Amiet's method⁵ extended to rotating frames,⁶ coupled with a wall-pressure spectrum (WPS) model have been widely used in propeller and wind turbine aeroacoustic community to enable rapid characterization of broadband noise level of rotating blades. Such methods have been shown to deliver adequate accuracy for propellers and rotors operating at high advance ratios, near the operating points of optimal aerodynamic efficiency.⁷ However, due to their inherent level of empiricism and modelling assumptions, the accuracy of these methods drastically deteriorates as portions of the blades enter stall at lower advance ratios. Large variability in predicted noise level and lack of robustness are also observed when different underlying WPS models are used.⁸

In this work, we seek to circumvent this predicament by exploring a data-driven framework where a predictive machine learning (ML) model is trained using a large number of relatively inexpensive low-fidelity simulations and enhanced by a small number of high-fidelity measurements obtained from wind-tunnel experiments. In the deployment stage, this ML model can be used to predict the acoustic (farfield noise level) performances of isolated propellers in forward flight, and each prediction only requires an evaluation of the ML model which is usually extremely fast. Specifically, we will use techniques of transfer learning (TL)⁹ to incorporate information from multiple fidelity of data sources. While ideally the ML model is trained using ample high-fidelity experiment data, they are usually limited in number since wind tunnel experiments are expensive and time-consuming. Instead, an initial baseline ML model will be built using a relatively large, accessible data set composed of only inexpensive low-fidelity simulations, and then fine-tuned with just a few high-fidelity wind tunnel experiment measurements. Hence, TL combines the power of *number* from the low-fidelity data set with *fidelity* from the high-fidelity experiment data set, to produce an overall model that achieves better predictive performance than if only the low-fidelity or only the high-fidelity data set is used alone.

For low-fidelity simulations, we use a semi-analytical Amiet-type airfoil trailing-edge noise (TEN) model extended to rotating blades⁶ for broadband noise prediction. The time-to-solution of the low-fidelity predictions is several orders of magnitude lower than that of high-fidelity scale-resolving simulations and hence allows for a large number of low-fidelity simulations to be performed for the initial ML training. The high-fidelity experiments have been performed in the aeroacoustic wind tunnel at the University of Bristol using 3D printed propeller blades.

The remainder of this paper is organized as follows: various components of the ML-based propeller prediction framework including the low-fidelity simulation method, experimental set-up, transfer learning approaches are described in Section II. The results are presented in Section III, while the conclusion and outlook to the final paper are outlined in Section IV.

II. Methodology

The overall framework of the multi-fidelity data-driven model for propeller noise prediction is developed as follows: the definition of propeller design space and the selection of input features for the machine learning study is first determined in Section II.A. A large number of randomly generated samples within this design space are then simulated using the low-fidelity propeller noise prediction tool presented in Section II.B. The details of the high-fidelity aeroacoustic experiment, including blade design, manufacturing and wind tunnel measurements are discussed in Section II.C.2. The baseline machine learning model trained initially with only the low-fidelity data is enriched by the

selected experimental data at different operating conditions (*i.e.*, propeller rotational speeds and free-stream velocities) using the transfer learning technique outlined in Section II.D.

II.A. Definition of Propeller Design Space

For this study, we focus only on two-bladed propellers with a fixed radius of $D = 22\text{cm}$. Both the blade twist and cross-sectional profile have a strong influence on the aerodynamic performance and noise signature of an isolated propeller and are therefore allowed to vary. Specifically, blade twist was varied via the pitch-to-diameter ratio (P/D), while the cross-sectional profiles of the propeller blades were controlled by the three parameters which subsequently define the conventional NACA 4-digit airfoil profiles, namely maximum camber (M_c), location of maximum camber (C_{max}) and maximum thickness (t_c). To allow the examination of ‘only’ the effects of the airfoil profiles, no level of taper was introduced to any of the blades. Thus, all blades have a constant airfoil profile over the full span of the blades, with the profile maintaining a constant chord length of $c = 2.3\text{cm}$, corresponding to a constant chord-to-radius ratio of c/R of 0.2. Figure 1 shows the blade angle (β) distribution and the chord length along the spanwise direction for two propeller cases with of $P/D = 0.8$ and 1.2, respectively, which are the lower and upper bounds of the P/D ratios used in the present study. The tip of each blade was kept square. In addition, for each propeller, the free-stream velocity (U_∞) and propeller rotational speed (Ω) are both considered design variables, while the incoming flow is laminar, parallel to the rotation axis and uniform across the propeller plane of rotation.

In summary, the propeller design variables selected for this study consist of four propeller shape variables (M_c , C_{max} , t_c , and P/D) and two operating conditions (Ω and U_∞). These constitute the six inputs to the ML model. Table 1 shows the upper and lower bounds for each of these variables. Furthermore, it was observed in experimental studies that as the flow speeds is increased above 10m/s, the propeller tends to start windmilling at lower RPMs and partially stalling at relatively high RPMs. To avoid these undesirable off-design conditions, the design space is further restricted as shown in Figure 2, similar to that of the previous work by the authors.¹⁰

For the present study, both numerical and experimental sound pressure level (SPL) results from randomly selected propellers within the design space are used to train and enrich the ML model, as such to assess the ability of the trained neural network to accurately predict the broadband noise component of the propeller far-field acoustics. The broadband frequency range is limited $1000 \leq f \leq 7000$. In addition to the noise spectra, results from three polar angles of 45° , 90° and 135° polar angles were extracted to help examine whether the trained model is able to capture the directivity behaviour of the propeller noise, which is also essential in understanding and predicting propeller noise.

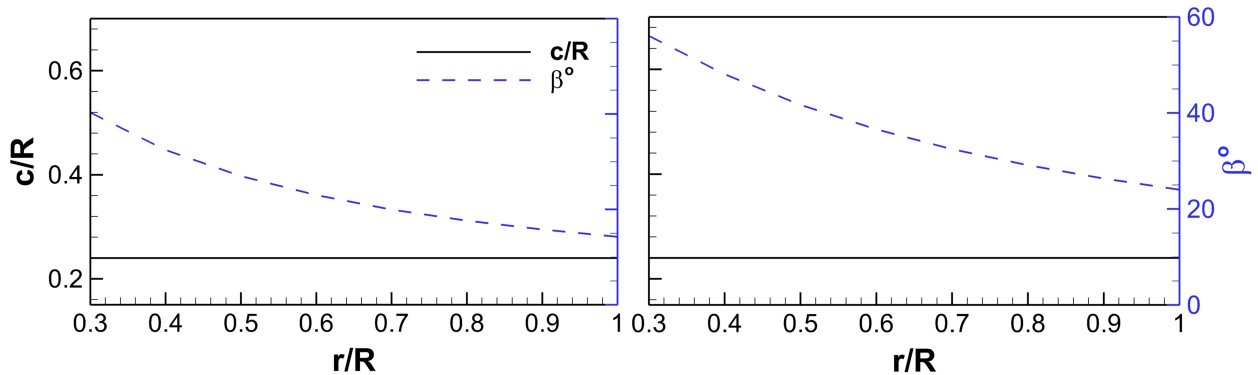


Figure 1: Spanwise twist angle distributions and chord lengths for the blade with $P/D = 0.8$ and $P/D = 1.4$, representative of the lower and upper pitch-to-diameter bounds.

II.B. Low-Fidelity Propeller Simulation

II.B.1. Propeller Load Simulation with BEMT

For predicting the load produced by a propeller in forward flight, Blade Element Momentum Theory (BEMT) is used. BEMT is a combination of two widely known analytical methods for estimating propeller loads: blade element theory and momentum theory. Blade element theory splits the propeller blade along the radial direction into a number of 2D sections. The forces on each section can then be obtained through airfoil theory and integrated along the blade to produce the propeller thrust and torque. This method is combined with momentum theory, which treats the propeller as an actuator disk. By combining these two methods the induction of the freestream fluid by the propeller into

Design Variable		Lower Bound	Upper Bound
Maximum Camber (% chord)	M_c	0	5.0
Location of Maximum Camber (% chord)	C_{max}	20.0	50.0
Thickness (% chord)	t_c	10.0	20.0
Pitch-to-Diameter Ratio	P/D	0.8	1.4
Rotational Speed (RPM)	Ω	3000	6000
Flow Speed (m/s)	U_∞	8.0	20.0

Table 1: Propeller design variables and their bounds considered in this study, following that of previous work¹⁰

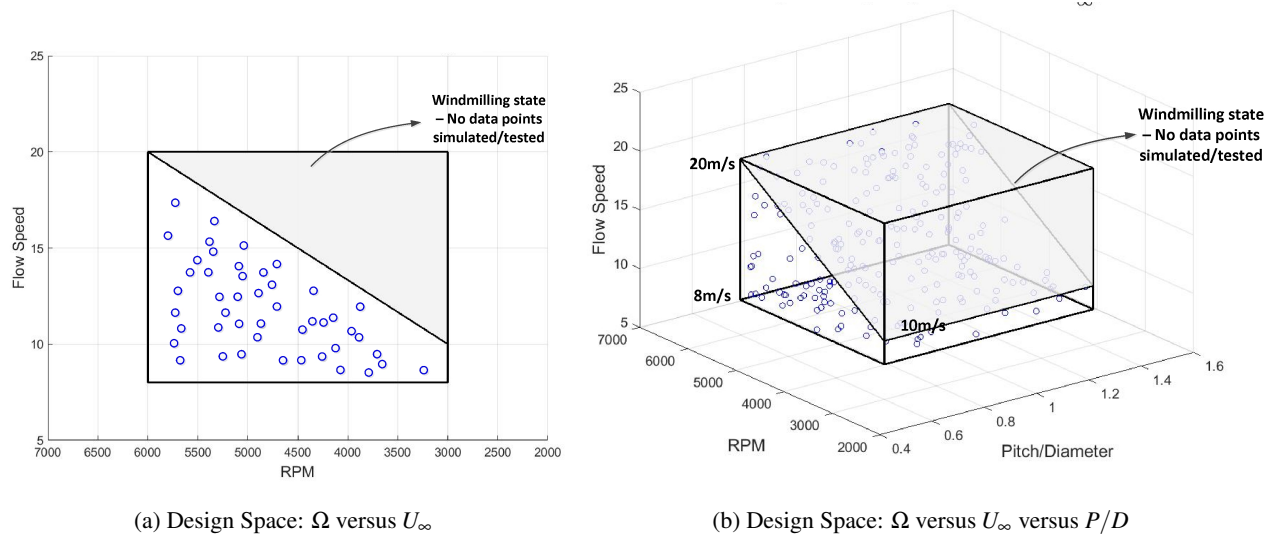


Figure 2: Further restriction of propeller design space to avoid wind-milling and stall (grey region), similar to the previous work¹⁰

the rotor plane is taken into account when estimating the 2D sectional loads, producing a more accurate prediction. Spanwise flow effects along the blade are not modelled by BEMT, so a Prandtl tip loss function is included in the implementation to better predict the reduction in sectional lift near the tip and hub. The aerofoil load estimation is performed using a deep neural network that was trained using a large set of XFOIL predictions. More details on this BEMT implementation are provided by Pullin *et al.*¹¹

II.B.2. Broadband Noise Prediction using Amiet's Trailing-Edge Noise Model

The trailing edge noise component of propeller broadband noise is generated by the interaction of turbulent pressure fluctuations in the boundary layer of the blade interacting with the trailing edge. Directly modelling the turbulent structures in the boundary layer requires expensive and unsteady fluid simulations. Empirically based methods can be used to estimate these pressure fluctuations and the resulting far-field acoustics. The frequency domain formulation by Amiet⁵ is a well known semi-analytical method for predicting the trailing edge noise spectrum generated by an aerofoil. This method uses as a key input parameter the Wall Pressure Spectrum (WPS) at the trailing edge, which is determined in this work through the use of a semi-empirical model by Schlinker and Amiet.¹² This model uses the boundary layer displacement thickness and edge velocity near the trailing edge to estimate the WPS.

To extend the airfoil noise to a rotor a formulation by⁶ is used. First the blades are divided into strips and the propeller noise at a farfield observer location is determined by a summation of the individual section contributions. This is then integrated over the azimuthal coordinate of the blade with a correction applied for Doppler effects occurring due to the rotation of the blades. The local flow conditions for each section strip along the propeller blade is determined through the BEMT method outlined previously in this section. The deep neural network used to predict the aerofoil load estimation for that method is also used to provide the boundary layer inputs based on XFOIL training data, which is then supplied to the WPS model. As such, the blade discretization used for the BEMT method is preserved for the

trailing edge noise calculation.

II.C. Experimental Set-up

II.C.1. Blade Design and Manufacturing

Since there are a number of parameters influencing the aerodynamic performance and noise signature of an isolated propeller. As a proof-of-concept study, parametric variations of the propeller blades are limited to the aerofoil profiles and the pitch-to-diameter ratios, as specified in Sec. II.A.

The propeller blades were manufactured using additive manufacturing techniques with a resin based stereolithography apparatus (SLA) printer, which has been previously verified to produce propellers with accurate profile resolution, sufficiently smooth surface finish and material strength at moderate rotational speed. The SLA prints were oriented such that the layers of plastic were printed in the radial direction towards the tip of each blade, achieving an improved surface finish with minimal support material. By printing in the radial direction, each propeller blade maintained a smooth airfoil profile throughout its length. The blades were then fitted to a propeller hub manufactured by varioPROP,¹³ making the changes between different blades into a two-bladed propeller assembly relatively straightforward. An example of one of the SLA-printed blades fitted to the varioPROP hub is shown in the inset of Figure 3.

II.C.2. Experimental Condition and Measurements

To obtain high-fidelity far-field noise data needed for enriching the ML model, acoustic experiments on isolated propellers were conducted at the aeroacoustic wind tunnel facility at University of Bristol. The closed-loop open jet facility is anechoic down to 160Hz. The jet nozzle exit dimension of 0.775m (H) $\times$ 0.5m (W), which yields a blockage factor of less than 10% with the propeller blades. The free-stream velocity was varied from $U_\infty = 8\text{m/s}$ to 20m/s at 4m/s interval, to investigate the effects of advance ratios. Here, the advance ratio, J , is defined as $J = U_\infty / (nD)$, where $n = \Omega/60$ is the rotational speed in revolution per second. The rotational speed of the propeller was controlled through an electronic speed controller (ESC) to vary from 3000 to 6000RPM (revolution-per-minute). The laminar free-stream has an unfiltered turbulence intensity of less than 0.2% across the velocity range tested, hence no turbulence interaction noise is expected. Readers are advised to refer Mayer et al. for more details on the facility.¹⁴

Figure 3 provides an schematic view of the isolated propeller set-up, which has been used for several other studies.^{15,16} To obtain the far-field spectra and directivity pattern, an overhead array consisting of 23 GRAS 40PL free-field microphones was used. The semi-circular array spans a polar angle from 40° to 150°, which a smaller polar angle indicates an upstream direction. Signals from all 23 microphones were simultaneously captured and registered using National Instruments sound and vibration acquisition system. To achieve statistical convergence, acoustic data were collected for a duration of 16s at sampling rate of 2¹⁶Hz, and subsequently were subjected to Welch's method in to derive the sound pressure level (SPL) of the pressure fluctuations in frequency domain. The SPL of the far-field noise was determined as $10\log_{10}(\phi_{pp}\Delta f/p_0^2)$, where ϕ_{pp} is magnitude squared acoustic pressure fluctuations and p_0 denotes the reference pressure of 20 μ Pa. The frequency resolution (Δf) of the Welch's analysis was evaluated at 1Hz in order to directly compare with results from the numerical simulations.

II.D. Transfer Learning from Low-Fidelity Simulations

We adopt deep neural network (DNN) as our ML model for its expressiveness. Our overall goal is to train a DNN that maps from six design features $x = \{M_c, C_{max}, t_c, P/D, \Omega, U_\infty\}$ to acoustic SPL in the high-fidelity experiments at three polar angles 45°, 90°, 135° (see Figure 3 for the polar angle definitions) each at 7 frequencies 1000, 2000, ..., 7000 Hz, for a total of 21 output targets denoted collectively by y . The DNN prediction is then

$$\hat{y} = f(x; w), \quad (1)$$

where $\hat{y}$ denotes the DNN *prediction* of the output quantities, and w collects all tunable DNN parameters (i.e. weight and bias parameters). A depiction of the DNN architecture is shown in Figure 4. Given a training data set $\mathcal{D} = \{x_n, y_n\}_{n=1}^N$, the DNN is trained by finding the optimal w^* that minimizes a loss function $\mathcal{L}$ (e.g., mean squared error):

$$w^* = \arg \min_w \mathcal{L}(w; \mathcal{D}) = \arg \min_w \left\{ \frac{1}{N} \sum_{n=1}^N [f(x_n; w) - y_n]^2 \right\}. \quad (2)$$

However, the expensive experiment data is scarce and too few to achieve good training alone. Therefore we leverage information from low-fidelity simulations in order to supplement the high-fidelity data through the process of TL.

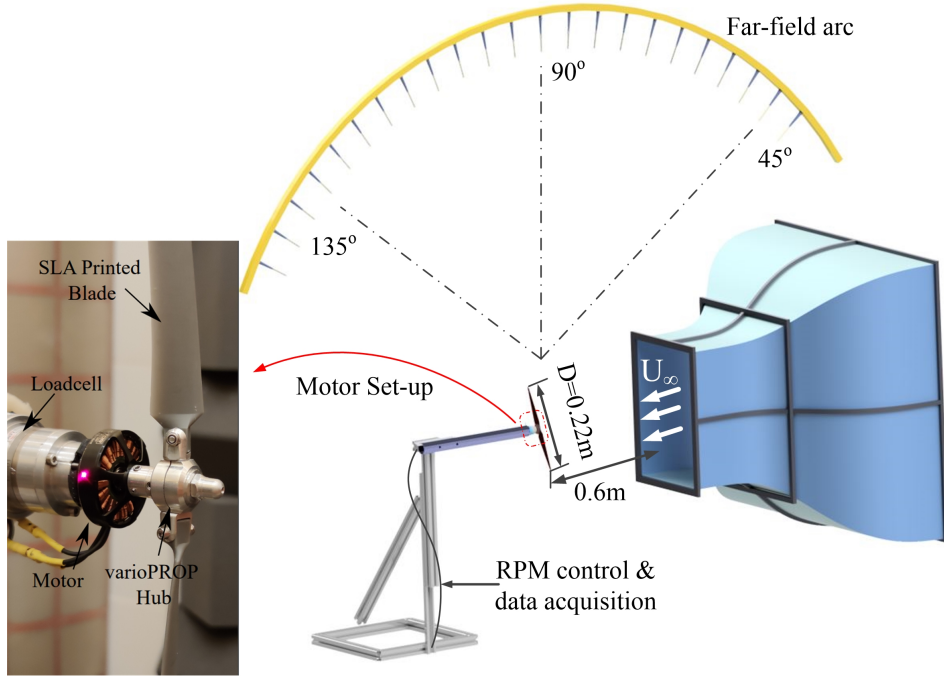


Figure 3: Schematic of the experimental set-up of the far-field acoustic measurements of an isolated propeller.

The simplest implementation of TL begins by training a model on (typically abundant) data from a *source domain*, in our case that is the low-fidelity simulation data set $\mathcal{D}_{LF}$. This allows us to obtain a DNN whose parameters are

$$w_{LF} = \arg \min_w \mathcal{L}(w; \mathcal{D}_{LF}). \quad (3)$$

The subsequent task is then to fine-tune to this model in order to adapt the model to a different, but related *target domain* where only a small amount of data is available,¹⁷ in our case that is the data set of high-fidelity experiments $\mathcal{D}_{HF}$. We then partition the overall parameters into two subsets $w = \{w_1, w_2\}$, for example where w_2 are the parameters of the last layer of the DNN while w_1 represents the rest. An effective TL technique then “freezes” w_1 to their corresponding values of $w_{LF,1}$, and only retrain w_2 using the new $\mathcal{D}_{HF}$ (illustrated in Figure 4):

$$w_2^* = \arg \min_{w_2} \mathcal{L}(\{w_{LF,1}, w_2\}; \mathcal{D}_{HF}). \quad (4)$$

Upon completion, the TL model has final parameter values $w = \{w_{LF,1}, w_2^*\}$.

Freezing the parameters from all but the last layer preserves many of the general trends that were learned from the related source domain while still allowing trends unique to the target domain to be learned and corrected. Another benefit of this TL approach is that the influence of parameter over-fitting is greatly mitigated, since the dimensionality of parameters only from the last layer is substantially smaller than from the entire DNN. Over-fitting is a major concern when learning with small data sets. Training only the last layer of parameters on the target data allows for (or be closer to) an over-determined system, whereas the DNN would otherwise likely be over-parameterized and under-determined if we trained all of its parameters (i.e. from all layers) on the small target data set.

III. Results

III.A. Low-Fidelity Propeller Performance Prediction and Training Data Generation

Low-fidelity propeller broadband noise predictions based on Amiet’s method with the Schlinker WPS model were performed on 54,000 propellers inside the design space illustrated in figure 2. This produced the set of low fidelity predictions for the broadband SPL at frequencies between 1kHz and 7kHz at intervals of 1kHz, for observer locations corresponding three polar angles ($R = 1.75m$, $\theta = 45^\circ, 90^\circ, 135^\circ$) in the farfield microphone array depicted on Figure. 3.

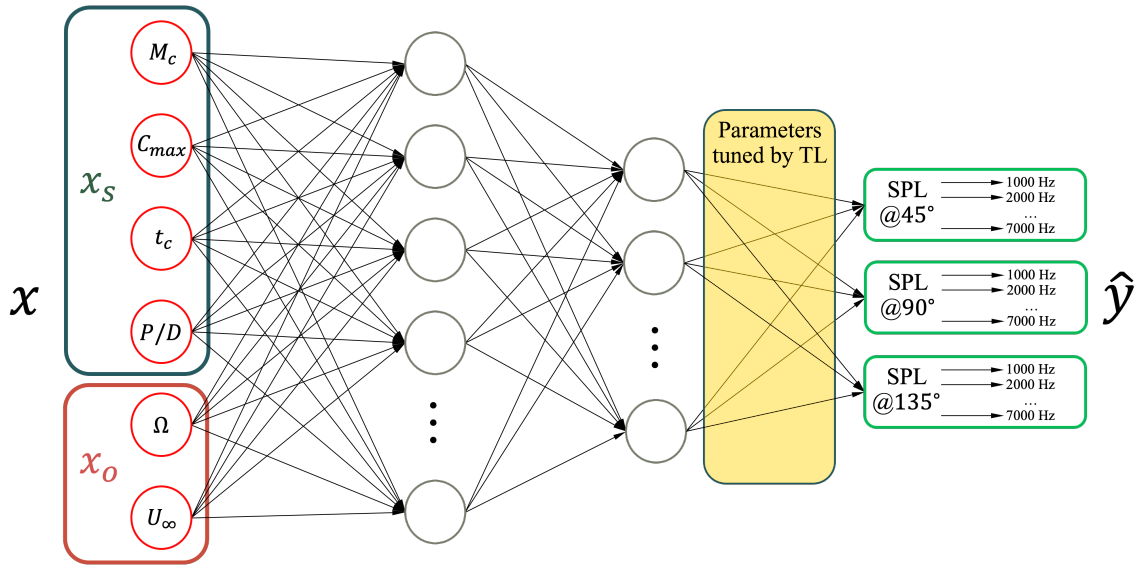
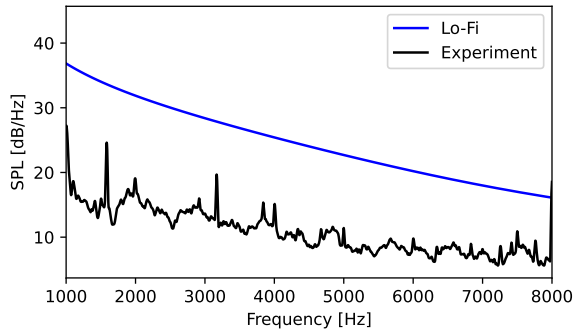


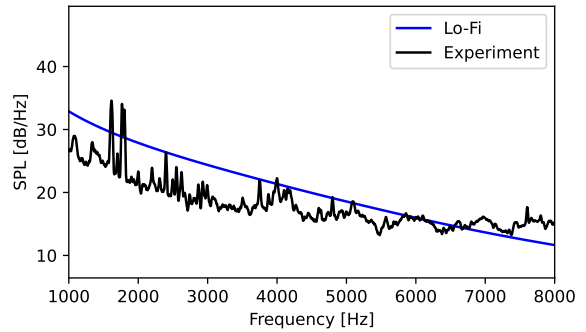
Figure 4: A DNN is first trained using source data set $\mathcal{D}_{LF}$. During TL, only the last layer's parameters (highlighted in orange) are fine tuned using the target data set $\mathcal{D}_{HF}$ while all other parameters are frozen at values obtained from the source data set. Note that the overall input vector x can be further grouped into shape parameters x_s and operating conditions x_o .

A comparison between low-fidelity prediction and experiment in the broadband range at in-plane microphone ($\theta = 90^\circ$) for two randomly selected propeller designs and operating conditions is shown in figure 5. Both cases are inside the design space defined in Figure 2. It is evident that the accuracy of the low-fidelity model varies drastically depending on the propeller geometry and operating conditions. In the case of the first propeller (Figure 5(a)), the discrepancy is as much as 20 dB/Hz. This discrepancy is largely due to the significant level of empiricism and assumptions made in both Amiet's trailing edge noise model and the underlying WPS model. This points to a severe limitation of the low-fidelity Amiet-based model and its lack of robustness in predicting propeller broadband noise, and underscores the importance of the proposed data-driven framework in which a machine learning model trained predominantly with low-fidelity results is enhanced by high-fidelity experimental measurements.

The main advantage of using this method for the generation of the baseline samples is its computational efficiency. The performance of the 54,000 low-fidelity samples used in this study were evaluated in minutes on consumer-grade hardware. In contrast, a single high fidelity simulation of propeller aerodynamics and acoustics such as LBM or LES coupled with an acoustic analogy would take many hours, even days on a HPC cluster to evaluate.



(a) NACA0012, $P/D=1.0$, 5000RPM, $U_\infty = 8.0m/s$



(b) NACA3320, $P/D=0.9$, 6000RPM, $U_\infty = 16.0m/s$

Figure 5: Comparison between low-fidelity prediction and experiment in the broadband range at in-plane microphone ($\theta = 90^\circ$) for two randomly selected points in the design space defined in Table 1

III.B. Experiments

The experiments were performed with a total of 18 randomly chosen blades to enrich the ML model and improve the prediction accuracy of the DNN. Two more blades were tested in the same round of measurement as validation to DNN performance. Table 2 populates the parameters of the blades used in the experiments. As can be seen, variations of the profiles and P/D ratios of the first 8 blades are more controlled than the remaining 10 blades. This is due to the fact that from the previous work,¹⁰ it was found that the prediction accuracy improves by training different blades occupying both the moderate (or well controlled) and the extreme (inclined towards higher P/D and max camber) parameter space.

Prior to undertaking the detailed analyses and training process for the ML models, it is useful to first understand the broadband noise behaviour of the tested blades under distinct operating conditions. Figure 6 shows the noise spectra for propellers with NACA 0012 profile and two different pitch-to-diameter ratios of $P/D = 0.8$ and 1.2 , respectively, at three free-stream velocities of $U_\infty = 0, 8$ and 16m/s and rotational speed of 5000RPM . The overall far-field acoustic characteristics of the isolated propeller consists of a series of tonal peaks at the blade pass frequency and its harmonics as well as notable broadband humps at the mid-to-high frequencies of approximately $f \geq 1000\text{Hz}$, typical of the noise spectra from an isolated propeller.^{3,16} Taking a closer look at the broadband components of the spectra, decreasing the free-stream velocity while maintaining the rotational speeds (thus reducing the advance ratio), as shown in Fig. 6(a), yields notable increase in the broadband noise over the frequency range of $800 \leq f \leq 8000$, suggesting strong correlations between the operating conditions and the broadband noise components. Furthermore, comparing two different P/D ratios of 0.8 (see Fig. 6(a)) and 1.2 (see Fig. 6(b)), blades with higher P/D ratios generally exhibit more pronounced broadband noise level, though similar trends can be observed when varying the advance ratio.

Moreover, when integrating over the broadband frequency range from $1000 \leq f \leq 7000\text{Hz}$, which is the frequency range of interest of the present prediction, the integrated sound pressure level (OASPL) indicates a clear directivity pattern of the broadband noise component with the lowest level directly above the propeller plane of rotation at $\theta = 90^\circ$ and gradually increasing towards shallow angles of $\theta = 45^\circ$ and 135° . Moreover, the OASPL results corroborate the observations from the noise spectra. The broadband noise component is significantly higher at low free-stream velocity as well as for larger P/D ratios. Clearly, the identified parameters from both the blades and the operating conditions are influential to the broadband noise performance and directivity behaviour of the isolated propellers. Hence, these parameters are considered the key inputs to the training process for the ML model and the accurate prediction of the broadband noise.

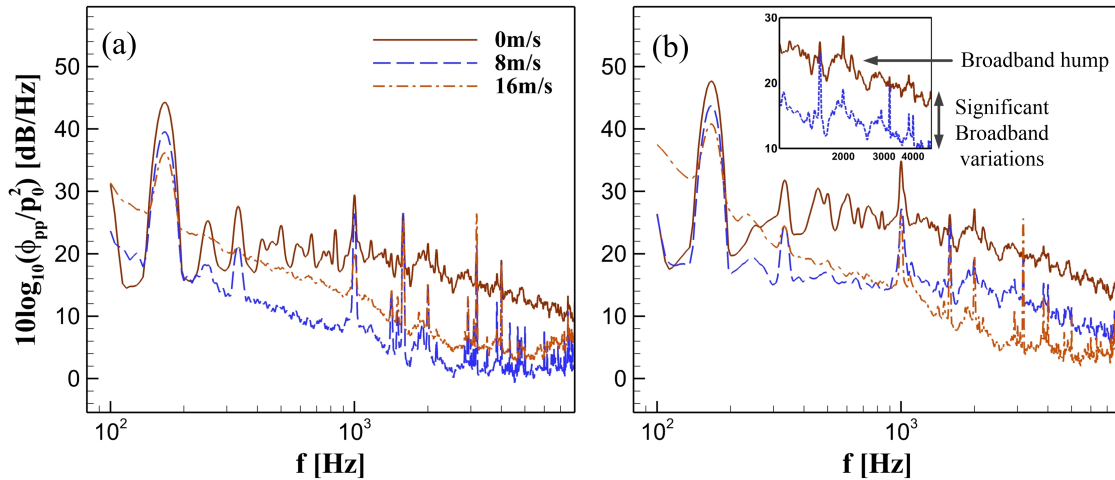


Figure 6: Comparison of the far-field noise power spectral density of the NACA 0012 propellers for (a) $P/D = 0.8$ and (b) $P/D = 1.2$ at rotational speed of 5000RPM .

III.C. Transfer Learning

We adopt a densely connected DNN architecture with two hidden layers, totaling around 1573 tunable parameters (Figure 4 depicts the architecture and Table 3 details the layer and parameters). All data sets involved are summarized in Table 4 with detailed blade shape parameters for each set captured in Table 2. A baseline DNN with all 19,605

Experimental campaign with randomly selected blades				
	Pitch to diameter ratio (P/D)	Max camber (M_c)	Max camber location (C_{max})	Thickness (t_c)
Blade1	0.8	0	0	12
Blade2	1.0	0	0	12
Blade3	1.2	0	0	12
Blade4	1.4	0	0	12
Blade5	0.8	4	4	12
Blade6	1.0	4	4	12
Blade7	1.2	4	4	12
Blade8	1.4	4	4	12
Blade9	1.4	4	5	12
Blade10	1.4	5	5	12
Blade11	1.4	3	5	12
Blade12	1.4	4	5	10
Blade13	1.4	4	5	14
Blade14	1.3	4	5	12
Blade15	0.9	0.5	2	18
Blade16	1.0	0.5	2.5	20
Blade17	0.9	0.5	3.25	20
Blade18	1.1	1.5	2.0	20
Validation experiments				
	Pitch to diameter ratio (P/D)	Max camber (M_c)	Max camber location (C_{max})	Thickness (t_c)
BladeV1	0.9	3	3	20
BladeV2	1.2	5	4	18

Table 2: Blade shape parameters used for both rounds of experiments and the validation cases (20 in total). Note that all parameters are expressed as percentage to the aerofoil chord of $c = 2.86\text{cm}$.

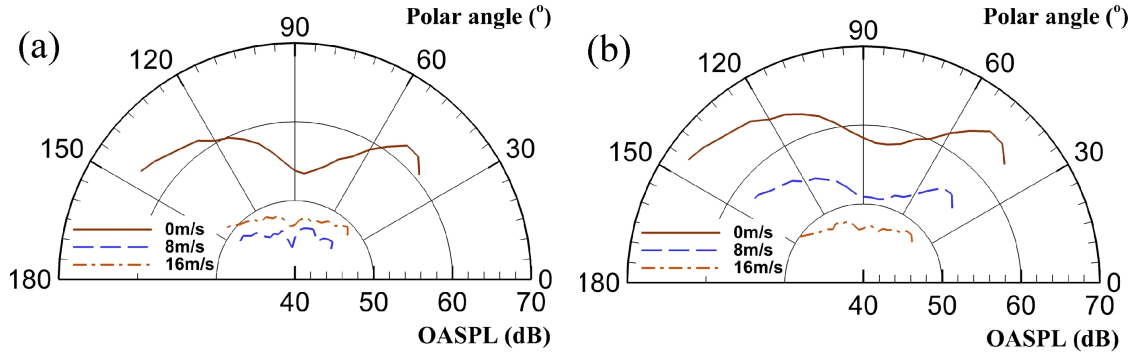


Figure 7: Comparison of the integrated sound pressure levels of the broadband components (from 1000 Hz to 7000 Hz) from the NACA 0012 propellers for (a) $P/D = 0.8$ and (b) $P/D = 1.2$ at rotational speed of 5000 RPM.

tunable parameters is first trained for 1000 epochs using a data set of 54,000 low-fidelity simulations $\mathcal{D}_{LF}$. This DNN has an RMSE of 0.76 and an R^2 score of 0.99 with the lo-fidelity data, making it an accurate neural surrogate.

Subsequently, when TL is deployed, parameters are frozen to these low-fidelity values in all except the output layer, and these 1365 parameters are retrained using the 108 high-fidelity training data points $\mathcal{D}_{HF}$ from the wind tunnel experiments. Overall, we will compare 3 different DNN models trained using: (1) low-fidelity data set only (Lo-Fi), (2) high-fidelity data set only (Hi-Fi), (3) both low- and high-fidelity data sets via TL (TL).

Layer name	Tunable Parameters (Baseline)	Tunable Parameters (Transfer Learning)
Input	0	0
Dense 1	1,792	0
Dense 2	16,448	0
Output	1,365	1,365
Total	19,605	1,365

Table 3: Architecture of the baseline and transfer learned DNN. In the baseline DNN, all layers' parameters are trainable. In the transfer learning model, all but the last layer's parameters are frozen after training on low-fidelity simulations, and the last layer's parameters are retrained based on experimental data.

Data set name	Data Entries	Unique Experimental Geometries
Low-fidelity simulations	54,000	n/a
Experimental train:	108	18
Experimental test	16	2

Table 4: Unique experimental geometries are characterized by a unique combination of the shape parameters: $M_c, C_{max}, t_c, P/D$. Data is collected for each shape design at multiple Ω and U_∞ operating conditions.

Figure 8 presents the parity plots showing prediction versus experimentally measured SPL values under different design inputs using two validation propellers (see bottom of Table 2), where predictions are produced and compared across different DNN models (trained using different combinations of data sets) in the subfigures. If a model makes accurate prediction, the plotted points would be more aligned with the diagonal line in the figures. The overall root mean squared error (RMSE) is also shown for each ML model in Figure 9.

From these results, we observe that the Lo-Fi baseline NN model has poor performance: its samples in the parity plots (left column in Figure 8) are far from the diagonal line, and the RMSE in Figure 9 are the highest. The Hi-Fi NN model, in comparison, yields parity plots (middle column in Figure 8) that much better hug the diagonal line, and achieves a 91.8% decrease in RMSE over the Lo-Fi NN model on average over all frequencies and angles. The TL NN model similarly yields good parity plots (right column in Figure 8) and further improves to a 94.5% decrease in RMSE over the Lo-Fi NN model on average over all frequencies and angles. For individual frequencies and observation angles, the TL NN model outperforms the Hi-Fi NN model on 18 of the 21 target quantities (i.e. combinations of

observation angle and frequency).

Focusing on a single test point corresponding to test propeller V1 (NACA3320 airfoil profile, $P/D = 0.9$, see Table 2) operating at 6000 RPM and $U_\infty = 12.0$ m/s, the TL DNN model appears to have significantly improved the prediction of the broadband SPL over the entire frequency range of interest between 1000 Hz and 7000 Hz, showing excellent agreement with experimental data (see Figure. 10).

IV. Conclusions and Future Work

This paper presents an exploratory effort in developing a data-driven framework to improve the prediction of propeller broadband noise. In this framework, a multi-fidelity machine learning (ML) model, based on the transfer learning (TL) techniques outlined in Section II.D, was first trained using a large number of computationally inexpensive low-fidelity simulations and then enhanced by a small number of high-fidelity wind tunnel measurements using TL. To that end, an initial deep neural network was built with 54,000 low-fidelity training samples based on Amiet's method with Schlinker's semi-empirical wall pressure spectrum model. Then a set of 18 propeller blade designs were tested in different operating conditions in the aeroacoustic wind tunnel facility at University of Bristol. The high-fidelity measurement data obtained from this wind tunnel testing were used to update the ML model using a TL approach. In the deployment stage, this ML model enables rapid prediction of the broadband noise level between 1kHz and 7kHz of isolated 2-bladed propellers at three farfield observer locations given the airfoil profile, pitch-to-diameter ratio as well as forward speed and rotational speed of the propeller.

To demonstrate the efficacy of this data-driven framework, two additional sets of propeller blades, whose design parameters were randomly selected from the design space, were manufactured, tested in the wind tunnel, and were subsequently held out from ML-TL training. The comparison between these held-out high-fidelity test samples and the ML predictions indicated that models trained on high-fidelity data (with or without the inclusion of samples from the low-fidelity model) dramatically outperformed the low-fidelity model, as demonstrated by $> 90\%$ reductions in the root mean square error (RMSE). This shortcoming of the low-fidelity model largely motivates our data-driven approach for noise prediction. However, when low-fidelity samples were included in the training database of the ML model in a TL approach, error was further reduced by 28% relative to the high-fidelity-only model. These results demonstrate the effectiveness of a multi-fidelity approach in noise prediction.

For future work, we will further enhance the multi-fidelity ML model with additional high-fidelity experimental data. Since these experiments are quite expensive, entailing production of propeller blade prototypes and expenses from reserving the wind tunnel facility, an active learning (AL) algorithm will be used to help intelligently identify new propeller geometries to be manufactured and tested that may offer the greatest expected improvement to the ML model across the entire design space. To that end, the AL algorithm will be built around a Gaussian process regression model of the DNN model discrepancy and will be used to guide the selection of new experiments in regions with high discrepancy. This approach has been explored by the authors in an earlier work focused on propeller *tonal* noise prediction.¹⁰ It is then natural to consider, as a next step, applying such combined TL-AL approach to develop a multi-fidelity data-driven model to enable efficient prediction of propeller noise encompassing both tonal and broadband components. Additional input features such as propeller diameter, blade count, non-uniform chord distribution will also be considered in order to widen the range of applicability of the machine learning model.

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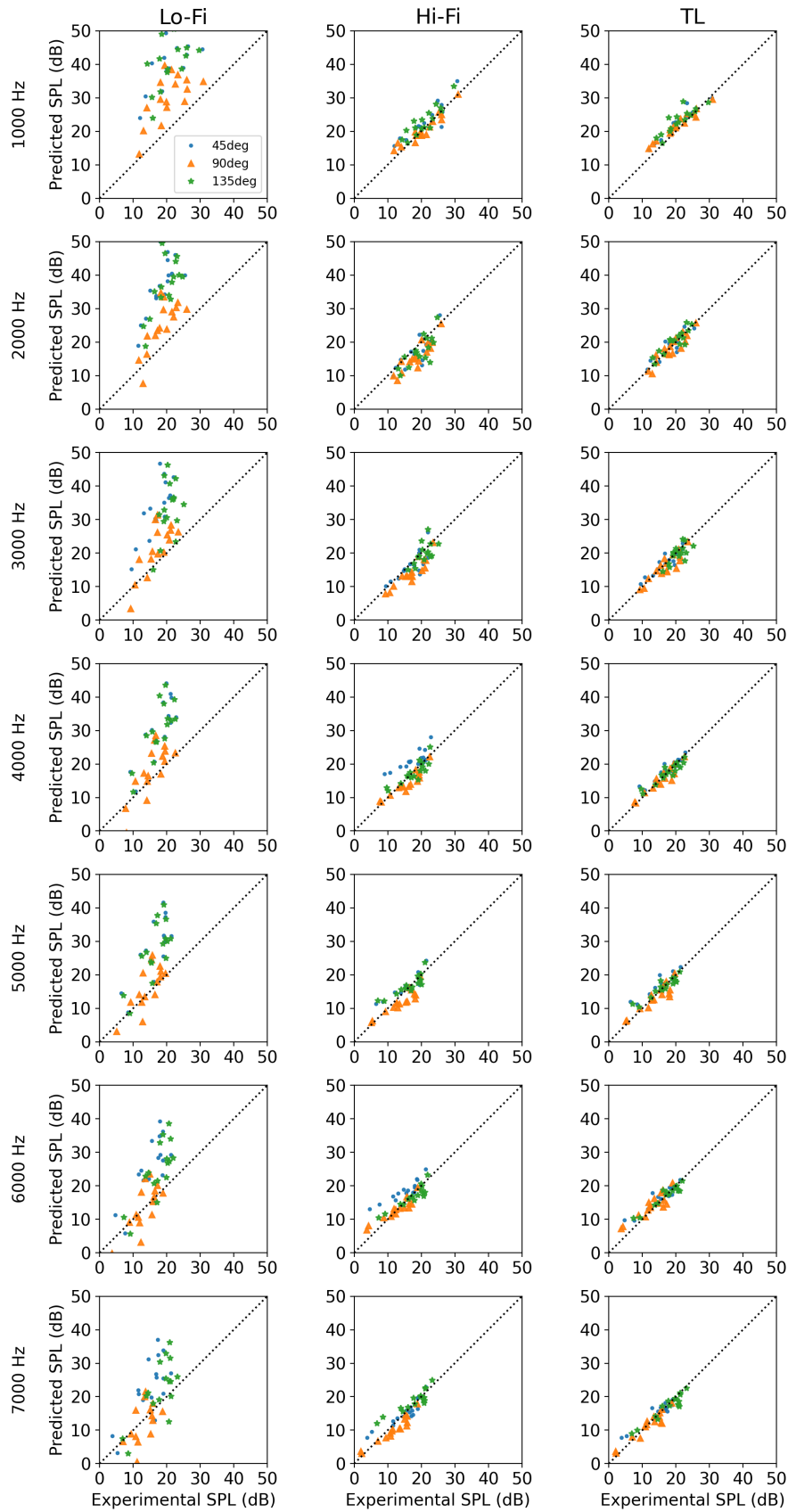


Figure 8: Parity plots comparing predicted versus experimentally measured SPL values of DNN at different observation angles and frequencies, using model trained with only low-fidelity data set (Lo-Fi, left column), only high-fidelity data set (Hi-Fi, center column), and both low- and high-fidelity dataset via transfer learning (TL, right column).

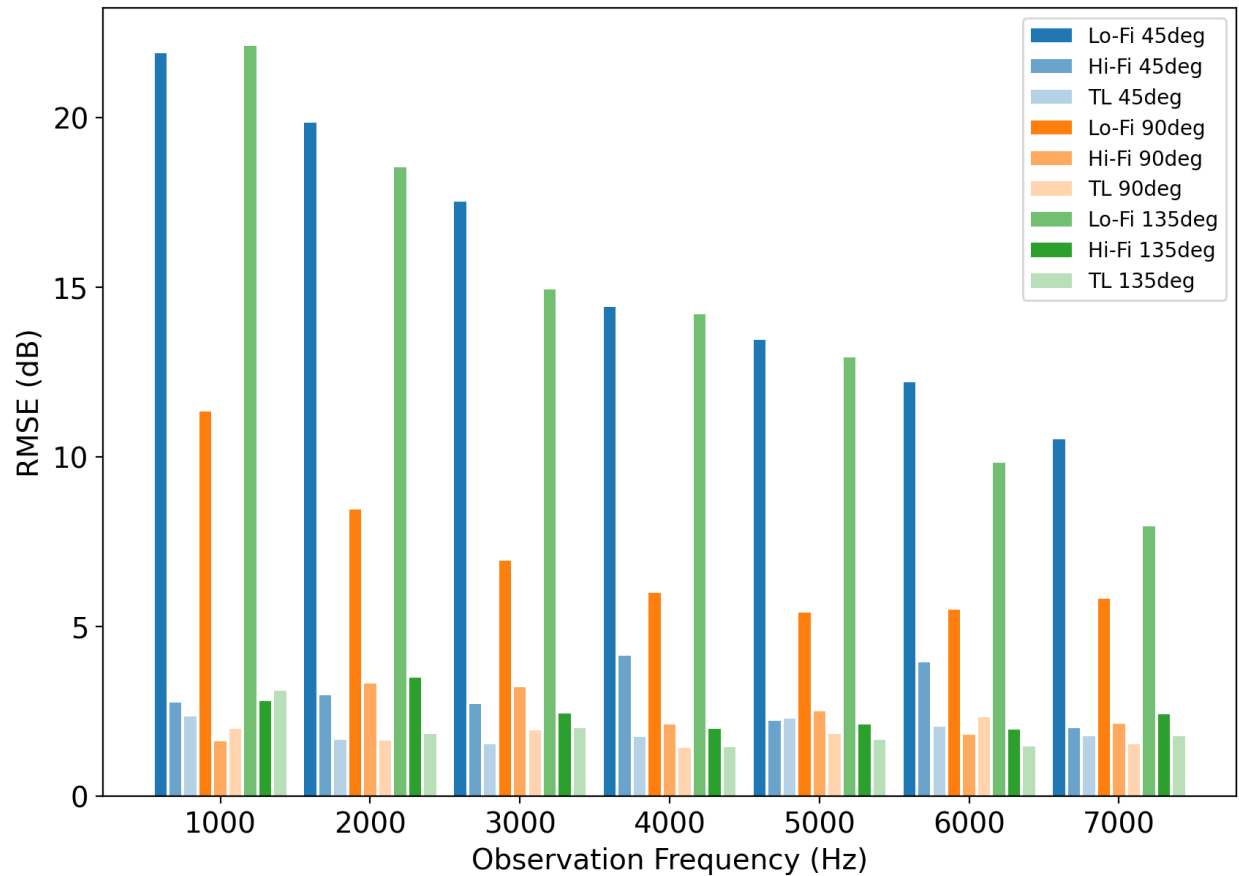
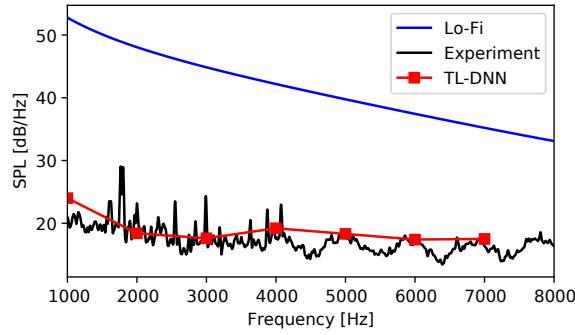
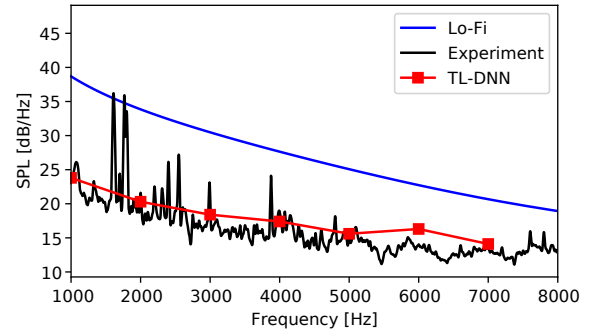


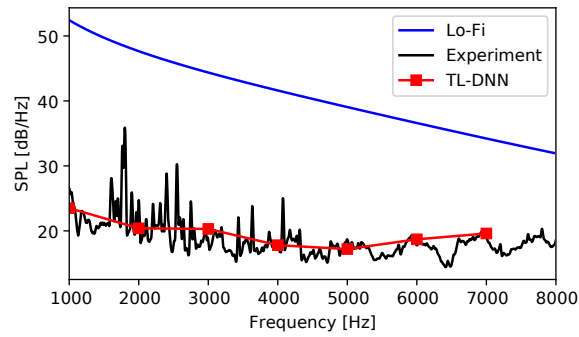
Figure 9: Comparison of root mean squared error (RMSE) of DNN at different observation angles and frequencies; models trained using only low-fidelity data set (Lo-Fi), using only high-fidelity data set (Hi-Fi), and using both low- and high-fidelity dataset via transfer learning (TL).



(a) $\theta = 45^\circ$



(b) $\theta = 90^\circ$



(c) $\theta = 135^\circ$

Figure 10: Comparison of broadband noise prediction between low-fidelity prediction, TL-DNN prediction and experimental data at three polar angles for test propeller V1 (NACA3320 airfoil profile, $P/D=0.9$, see Table 2) operating at 6000RPM and $U_\infty = 12.0m/s$.